

Wenjun Zhao

CONTACT INFORMATION

Department of Mathematics
Wake Forest University

zhaow@wfu.edu
<https://wenjunzhaowo.github.io>

EMPLOYMENT

NSF-Simons National Institute for Theory and Mathematics in Biology
Institute Affiliate Member, 2025-

Department of Mathematics, Wake Forest University

Tenure-track Assistant Professor, 2024-
Start date deferred by one year to hold PIMS-Simons Fellowship at UBC.

Department of Mathematics, University of British Columbia

PIMS-Simons Postdoctoral Fellow, 2024-2025
• Supervisors: Young-Heon Kim, Geoffrey Schiebinger, and Khanh Dao Duc

Division of Applied Mathematics, Brown University

LFZ Postdoc/Assistant Professor, 2021-2024
• Mentor: Bjorn Sandstede

EDUCATION

Courant Institute of Mathematical Sciences, New York University

Ph.D. , Department of Mathematics, 2021
• Advisor: Esteban G. Tabak

University of Science and Technology of China

B.S. , School of the Gifted Young, 2016
• Advisor: Yu-Hong Dai (Chinese Academy of Sciences)

INTERNSHIP & VISITING

Argonne National Laboratory, Mathematics and Computer Science Dept.

Wallace Givens Associate, 2020

University of Oxford, Department of Computer Science

Visiting student, 2015

RESEARCH INTERESTS

Optimal transport and distributional methods for modeling biological dynamics and shape variability.

PUBLICATIONS & PREPRINTS

(α - β): authors listed alphabetically.

PUBLISHED/ACCEPTED.

1. Zhao, W., Sutherland D.J., Dao Duc, K.,
Fast and Interpretable Quantification of Biological Shape Heterogeneity via Stratified Wasserstein Kernel,
PLOS Computational Biology (2026).<https://doi.org/10.1371/journal.pcbi.1014254>

2. Magana, S., Zhao, W., Dao Duc, K.,
Learning Continuous Morphological Trajectories via Latent Principal Curves,
Learning Meaningful Representations of Life (LMRL) Workshop at ICLR (2026).
3. Zhao, W., Fertig, E., Stein-O'Brien, G.,
CycleGRN: Inferring Gene Regulatory Networks from Cyclic Flow Dynamics in Single-Cell RNA-seq,
RECOMB (2026)., <https://www.biorxiv.org/content/10.1101/2025.11.12.688126v1>
4. Yu, S., Kushner, A., Teasell, E., Zhao, W., Srebnik, S., & Dao Duc, K.,
Advanced Coarse-grained Model for Fast Simulation of Nascent Polypeptide Chain Dynamics within the Ribosome,
Biophysical Journal (2026). <https://doi.org/10.1101/2025.04.08.647858>
5. Zhao, W., Larschan, E., Sandstede, B., & Singh, R.,
Optimal Transport Reveals Dynamic Gene Regulatory Networks via Gene Velocity Estimation,
PLOS Computational Biology (2025). <https://doi.org/10.1371/journal.pcbi.1012476>
6. Zhao, W., Maffa, S., & Sandstede, B.,
Data-driven Continuation of Patterns and their Bifurcations,
SIAM Journal on Applied Dynamical Systems (2025). <http://dx.doi.org/10.1137/24M165644X>
7. Zhao, W., & Tabak, E.G.,
Adaptive Kernel Conditional Density Estimation,
Information and Inference: A Journal of the IMA (2025). <https://doi.org/10.1093/imaiai/iaae037>
8. $(\alpha\text{-}\beta)$ Tabak, E.G., Trigila, G. & Zhao, W.,
The Conditional Barycenter Problem, its Data-Driven Formulation and its Solution through Normalizing Flows,
Communications in Mathematical Science (2024).
<https://dx.doi.org/10.4310/CMS.2024.v22.n6.a8>
9. $(\alpha\text{-}\beta)$ Zhang, H. & Zhao, W.,
A Memory-Efficient Neural Ordinary Differential Equation Framework Based on High-Level Adjoint Differentiation,
IEEE Transactions on Artificial Intelligence (2022). <https://doi.org/10.1109/TAI.2022.3230632>
10. $(\alpha\text{-}\beta)$ Tabak, E.G., Trigila, G. & Zhao, W.,
Distributional Barycenter Problem through Data-Driven Flows,
Pattern Recognition (2022). <https://doi.org/10.1016/j.patcog.2022.108795>.
11. $(\alpha\text{-}\beta)$ Tabak, E.G., Trigila, G. & Zhao, W.,
Data Driven Conditional Optimal Transport.
Machine Learning (2021). <https://doi.org/10.1007/s10994-021-06060-0>
12. $(\alpha\text{-}\beta)$ Tabak, E.G., Trigila, G. & Zhao, W.,
Conditional Density Estimation and Simulation through Optimal Transport,
Machine Learning (2020). <https://doi.org/10.1007/s10994-019-05866-3>.

PREPRINT.

1. Zhao, W., Fertig, E., Stein-O'Brien, G.,
Inferring Cyclic Dynamics from Unordered Single-Cell Snapshots via Invariant Measures,
Submitted to Cell Systems (as an extended version to RECOMB paper)
2. Chang, S., Zhao, W., Ma, Y., Sandstede, B., Singh, S.
Cell type-specific gene regulatory network inference from single cell transcriptomics with ctOTVelo
<https://www.biorxiv.org/content/10.64898/2026.03.11.711174v1>
Accepted as a poster at RECOMB.
3. $(\alpha\text{-}\beta)$ Tabak, E.G., Trigila, G. & Zhao, W.,
The Hierarchical Barycenter: Conditional Probability Simulation with Structured and Unobserved Covariates,
<https://arxiv.org/abs/2508.00206>.
In revision at Machine Learning.
4. $(\alpha\text{-}\beta)$ Lipnick, A.D., Tabak, E.G., Trigila, T., Wang, Y., Ye, X., Zhao, W.,
The Monge Optimal Transport Barycenter Problem,
<https://arxiv.org/abs/2507.03669>
In revision at International Statistical Review.
5. Zhao, W., Plaza-Rodriguez, A., Luanpaisanon, P., Wang, E.X., Gyllingberg, L., Singh, R., Fertig, E., Stein-O'Brien, G.,
Inferring the regulation dynamics of oscillatory networks from scRNA-seq data
<http://dx.doi.org/10.1101/2025.11.08.687360>
In review at Springer Lecture Notes special issue on "Computational Modeling of Complex Biological Systems: Collaborative Workshop for Women in Mathematical Biology".

TEACHING EXPERIENCE	Fall	2026	Instructor, Probability & Operations Research	Wake Forest
	Spring	2026	Instructor, Probability & Calculus II (Enrollment: 51)	Wake Forest
	Fall	2025	Instructor, Calculus II (Enrollment: 46)	Wake Forest
	Fall	2024	Small Class Instructor, Differential Calculus (Enrollment: 240)	UBC
	Spring	2024	Instructor, Essential Statistics (Enrollment: 95)	Brown
	Fall	2023	Instructor, Honors Statistical Inference I (Enrollment: 34)	Brown
	Summer	2023	Instructor, MATLAB mini-course for EDGE program	Brown
	Spring	2023	Instructor, Essential Statistics (Enrollment: 73)	Brown
	Fall	2022	Instructor, Honors Statistical Inference I (Enrollment: 84)	Brown
	Spring	2022	Instructor, Essential Statistics (Enrollment: 64)	Brown
	Fall	2021	Instructor, Statistical Inference I (Enrollment: 203)	Brown
	Spring	2021	Recitation leader, Intro to Fluid Dynamics, Complex Variables	NYU
	Fall	2020	Grader, Linear Algebra for Data Science	NYU
	Spring	2020	Recitation leader, Introduction to Math Modeling	NYU
	Fall	2019	Recitation leader, Introduction to Math Modeling	NYU
	Spring	2019	Recitation leader, Ordinary Differential Equations	NYU
	Fall	2018	Substitute lecturer/Grader, Partial Differential Equations	NYU
	Fall	2015	Teaching assistant, Multivariable Calculus	USTC

HONORS AND AWARDS	2025	Postdoc Teaching Prize in the Department of Mathematics, UBC
	2024	Rising Stars in Computational and Data Sciences, Oden Institute
	2024	PIMS-Simons Postdoctoral Fellowship
	2023	Dean’s Award for Excellence in Teaching, Brown
	2021	Named LFZ Assistant Professorship of Applied Mathematics, Brown
	2020	Nomination for Dean’s Dissertation Fellowship, NYU
	2019	NeurIPS travel award
	2019	Moses A. Greenfield Research Prize, NYU Courant
	2016–2021	Henry MacCracken Fellowship, NYU
	2015	Summer research fellowship at University of Oxford
	2015	Meritorious Winner in Mathematical Contest of Modeling
	2013	First prize in USTC Contest of Electromagnetics
	2012–2016	China National Encouragement Scholarship
SERVICES	• Leading organizer of NITMB workshop on “Representing and Learning morphology and shape dynamics from biological data” (2026) • Co-chair for publication session of ISCB on “Benchmarking machine learning publications: reporting, uncertainty and robustness”, program committee member of MLCSB track at ISMB (2026) • Leading organizer of SMB minisymposium on “Mathematical Methods for Biological Shape Analysis” (2025) • Co-organizer of Kantorovich Initiative Seminar at PIMS/UBC (2024-2025) • Mini-course instructor for EDGE (Enhancing Diversity in Graduate Education) (2023) • Co-organizer of Pattern Theory Group Seminar at Brown (2022-2024) • Ad hoc reviewer for: Nature Computational Sciences, Genome Medicine, Bulletin of Mathematical Biology, Pattern Recognition, Journal of Machine Learning for Modeling and Computing, Bioinformatics, Journal of Neuroscience Methods, Cell Systems, Mathematical Biosciences, Journal of Chemical Information and Modeling, Briefings in Bioinformatics, Computational Statistics and Data Analysis, RECOMB, ISMB, ACM-BCB, ICLR.	
CONFERENCES & WORKSHOPS	<i>TBD</i> (talk), IMSI Workshop on Statistical Foundations for the Analysis of Non-Euclidean Data, Chicago, US. (Oct 2026)	
	<i>TBD</i> (talk), ISMB 2026 MLCSB and SysReg tracks, DC, US. (July 2026)	
	<i>Learning Gene Regulatory Networks from Directed Cellular Dynamics</i> (talk), ASHBi Internation Workshop for Data Science X Mathematical Modeling, Kyoto University, Japan. (June 2026)	
	<i>Learning Gene Regulatory Networks from Directed Cellular Dynamics</i> (talk), Pacific Rim Conference of Mathematics, Academia Sinica, Taiwan. (June 2026)	
	<i>CycleGRN: Inferring Gene Regulatory Networks from Cyclic Flow Dynamics in Single-Cell RNA-seq</i> (talk), RECOMB, Thessaloniki, Greece. (May 2026)	
	<i>Quantifying variability in biological shape space through Wasserstein-type kernels</i> (talk), SIAM-NNP minisymposium on Optimal transport in biological sciences, State College, US. (Oct 2025)	

Dynamical GRN inference via optimal transport (talk), SMB minisymposium on Data-Informed differential equation modeling in cancer, Edmonton, Canada. (July 2025)

Data analysis through Wasserstein barycenter with general factors (talk), BIRS workshop on Wasserstein Gradient flows in Math and Machine Learning, Banff, Canada. (June 2025)

Data-Driven Methods for Two Inference Problems in Mathematical Biology via Optimal Transport (talk), ICERM workshop on UQ in mathematical biology, Providence, US. (May 2025)

Some metrics and statistics on quantification of shape variability (Talk), PIMS Postdoc Summit, Calgary, Canada. (Apr 2025)

Gene Regulatory Networks (GRNs) Inference Through Optimal Transport (Talk), SIAM Conference on the Life Sciences, Portland, US. (Jun 2024)

Data-driven quantification of patterns and their transitions (Talk), Boston University/Keio University/Tsinghua University workshop, Boston, US. (May 2024).

Data-driven methods for inference in dynamical systems with optimal transport (Talk), Rising Stars in Computational and Data Sciences, Oden Institute, US. (May 2024)

Quantifying patterns and their transitions in spatially extended systems (Poster), Dynamics Days, Davis, US. (Jan 2024)

Conditional optimal transport and its applications (Talk), Physical Applied Mathematics and Data Science, ShanghaiTech University, Shanghai, China. (Jan 2020)

Data Driven Conditional Optimal Transport (Poster), NeurIPS Optimal Transport in Machine Learning Workshop, Vancouver, Canada. (Dec 2019)

SEMINAR TALKS

TBD, Physics Colloquium, Wake Forest (Oct 8 2026)

Geometry and Dynamics of Biological Development from Unlabeled Point Clouds, CODES seminar, Emory (Apr 9 2026)

Invariant and Equivariant Data-driven Methods for Modeling Cellular Dynamics, Kantorovich Initiative Seminar, UBC (Nov 27 2025)

Invariant and Equivariant Data-driven Methods for Modeling Cellular Dynamics, Biomathematics seminar, NCSU (Nov 18 2025)

Learning Cellular Dynamics through Neural and Geometric Models of Imaging and Genomics Data, RealAI seminar, UCSB (Nov 7 2025)

Two Inference Problems in Dynamical Systems from Mathematical and Computational Biology, PIMS Emergent Research Seminar (Mar 26 2025)

Data-driven Continuation of Patterns and their Bifurcations, Department Colloquium, Western Washington University (Feb 13 2025)

Data-driven Continuation of Patterns and their Bifurcations, MSG seminar, NYU Courant (Jan 30 2025)

Data-driven methods for inference in dynamical systems via optimal transport, Level Set Seminar, UCLA (May 22 2024)

Optimal transport with covariates: Wasserstein barycenter and its extensions, Applied and Computational Math Seminar, Dartmouth College (March 28 2024)

Wasserstein barycenter for conditional density estimation and simulation, Computational and Applied Math Seminar, Tufts University (May 1 2023)

Data-driven Wasserstein barycenter problem, Leslie Comrie Seminar Series, University of Greenwich (Mar 30 2022)

Optimal transport and beyond, Math Slam, Brown University (Dec 2 2021)

Data-driven Wasserstein barycenter problem, LCDS & Pattern theory seminar, Brown University. (Oct 4 2021)

Optimal transport with covariates and its applications, APMA colloquium, Brown University. (Sept 23 2021)

Barycentric Optimal Transport: algorithms and applications, CAOS student seminar, New York University. (Nov 2020)

Advanced Neural ODE Solver through PETSc, Summer Argonne Students' Symposium 2020, Argonne National Laboratory. (Apr 2020)

Conditional optimal transport and its applications, CAOS student seminar, New York University. (Nov 2019)

A simplified entrainment model based on shallow water equation, CAOS student seminar, New York University. (Nov 2018)

Conditional density estimation through optimal transport, CAOS student seminar, New York University. (Dec 2017)

ADDITIONAL TRAINING

Rossbypalooza, University of Chicago, Chicago, USA. (July 2024)

Launch Course Design Institute, Sheridan Center for Teaching and Learning, Brown University, Providence, USA. (August 2021)

Science Communications Workshop, Arthur L. Carter Journalism Institute, New York University, New York, USA. (Oct 2019)

NASA JPL-Caltech Summer School: Using Satellite Observations to Advance Climate Models, Pasadena, USA. (Aug 2018)

REFERENCES

- Professor Bjorn Sandstede (Research, postdoc mentor)
Division of Applied Mathematics, Brown University,
bjorn.sandstede@brown.edu
- Professor Young-Heon Kim (Research, postdoc mentor)
Department of Mathematics, UBC
yhkim@math.ubc.ca

- Professor Matthew Harrison (Teaching)
Division of Applied Mathematics, Brown University,
matthew_harrison@brown.edu
- Professor Ritambhara Singh (Research)
Department of Computer Science, Brown University,
ritambhara@brown.edu
- Professor Esteban G. Tabak (Research, PhD advisor)
Department of Mathematics, New York University,
tabak@cims.nyu.edu